CEFUROXIME

Take tablets with food

Cross-allergy

CHEMICAL NAME (6R,7R)-3-(carbamoyloxymethyl)-7-[[(2Z)-2-(furan-2-yl)-2-methoxyiminoacetyl]amino]-8-oxo-5-thia-1-azabicyclo[4.2.0]oct-2-ene-2-carboxylic acid (sodium; axetil ester orally)

CLASS Second-generation cephalosporin

TRADE NAMES Zinnat (oral) · Zinacef (IV) · Ceftin · Cefuroxime axetil

THERAPEUTIC USES / INDICATIONS

- Respiratory infections: sinusitis, otitis media, bronchitis, community-acquired pneumonia.
- Urinary tract infections; uncomplicated gonorrhoea (historic).
- Skin and soft-tissue infections; Lyme disease.
- Surgical prophylaxis (IV).

ADMINISTRATION

- **PO** (cefuroxime axetil) with food — improves absorption.
- **IV:** slow injection over 3–5 min or infusion; **IM** deep.
- Tablets have bitter taste — swallow whole.

DOSAGES

- **Adult PO:** 250–500 mg every 12 h.
- **Adult IV:** 750 mg–1.5 g every 8 h (severe: 1.5 g every 6 h).
- **Children PO:** 10–15 mg/kg every 12 h (max 250 mg/dose); **IV:** 30–100 mg/kg/day in 3–4 doses (60 mg/kg/day for most infections).
- Reduce in CrCl <30 mL/min.

PHARMACOKINETICS

- **Peak:** PO 2–3 h. Oral bioavailability ~50% with food.
- Axetil ester hydrolysed in gut wall to active drug.
- Excreted unchanged in urine. **t½:** 1.2 h.

PHARMACODYNAMICS

- Inhibits **cell-wall synthesis** → bactericidal.
- Better Gram-negative cover than 1st generation: *H. influenzae*, *M. catarrhalis*, *E. coli*; stable to many β-lactamases.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea, nausea.
- Headache, dizziness.
- Rash; injection-site reactions.

ADVERSE EFFECTS (SERIOUS)

- Anaphylaxis.
- *C. difficile* colitis.
- Positive Coombs test, haemolytic anaemia (rare).
- Jarisch–Herxheimer reaction (Lyme).

CONTRAINDICATIONS & CAUTIONS

- Cephalosporin hypersensitivity; severe penicillin allergy.
- **Caution:** renal impairment, GI disease (colitis history).

KEY INTERACTIONS

- **Antacids, H2 blockers, PPIs:** ↓ absorption of oral cefuroxime.
- **Oral contraceptives:** possible ↓ efficacy.
- **Probenecid:** ↑ levels.

NURSING MANAGEMENT

- Check allergy history.
- Give tablets with food.
- Monitor clinical response and temperature.
- Monitor stools for diarrhoea.
- Renal function in elderly or long courses.

PATIENT EDUCATION

- Take with food every 12 hours.
- Complete the course.
- Report rash, breathing difficulty or severe diarrhoea.

KEY POINTS

- 2nd-gen cephalosporin.
- Oral axetil — take with food.
- Covers H. influenzae and Moraxella.
- Respiratory infections.

SOURCES CHECKED **[130]** Cefuroxime 1.5 g Powder for solution for injection vial (UK SmPC) **[131]** Cefuroxime 250mg tablets (UK SmPC) *Full references p. 432–457*